Composition of blood

COMPOSITION OF BLOOD (sample notes)

Blood is about 55% plasma and 45% formed elements (by volume, in typical adults, haematocrit around 45%).

Plasma
- About 90% water. Major proteins: albumin (oncotic pressure, transport), globulins (immunoglobulins among them), fibrinogen (clotting).

Formed elements
- Erythrocytes (red cells): anucleate, biconcave, carry haemoglobin. Lifespan about 120 days.
- Leukocytes (white cells): neutrophils, lymphocytes, monocytes, eosinophils, basophils.
- Platelets (thrombocytes): cell fragments from megakaryocytes; essential for primary haemostasis.

Clinical link: a full blood count (FBC) quantifies these elements and their indices (for example, MCV, MCH), giving first-line clues about anaemia, infection and bleeding disorders.